ALT2 – Brief

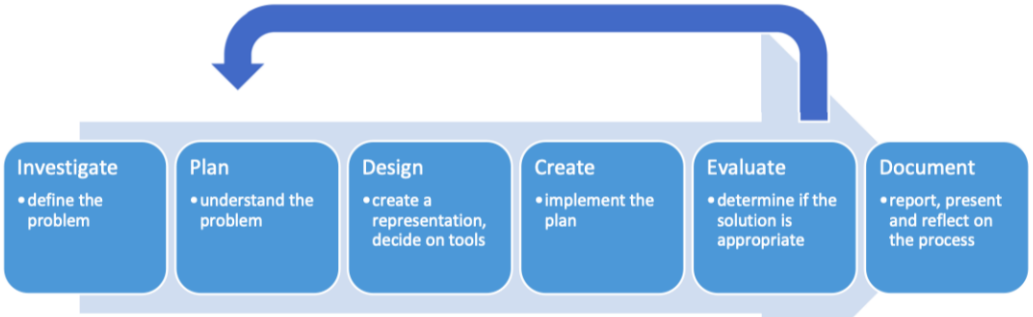

Figure 1. An iterative design cycle.
DES/NCCA (2018) *Leaving Certificate Computer Science curriculum specification*

This is our guide, our map and our dictionary. It tells us both what the department wants and how they want it said

Your Task

1. Come up with a Hypothesis to investigate using a dataset
2. Use python coding language to read in a dataset of your choice
3. Use python to prepare or examine raw data. You may need to edit the dataset to make it usable
4. Analyse this data to find:
    a. Mean
    b. Mode
    c. Median
    d. Any other you deem appropriate (range, interquartile range, standard deviation etc.)
5. Use python to create an appropriate graph to show trends.
6. State whether your hypothesis is true or false based on the analysis of the data

Groups

The project will be completed in groups of 3 or 4.

Deadline

The deadline for the project is Wednesday 8th May

Documents

When handing up your work you will create a zip folder with 2 folders inside it.

Folder 1: This folder contains all python files, csv files etc that you used during the project

Folder 2: This folder contains the project write up in html

<u>Project Write Up</u>

The website will have 4 webpages.

1. **Investigation and Plan (20 marks)**
2. **Design (20 marks)**
3. **Implementation and Testing (40 marks)**
4. **Evaluation (20 marks)**

1. **Investigation and Plan** (70-100 words)
   - State your ALT2 hypothesis
   - Evidence of research
     - Why did you choose this topic?
     - Where did you get your dataset?
2. **Design** (100 words)
   - Create a flowchart and explain each of the stages of the flowchart

3. **Implementation and Testing** (400 words)
   - Diary entries of daily progress
   - Explain why you made the decisions you made
   - Include screenshots of the code as it progresses through the project

4. **Evaluation** (150 words)
   - Did your hypothesis prove to be true or false?
   - Does your code help you answer the hypothesis?
   - What improvements or changes would you make if you had to do the project again?
   - What learning will you take forward into your next ALT?